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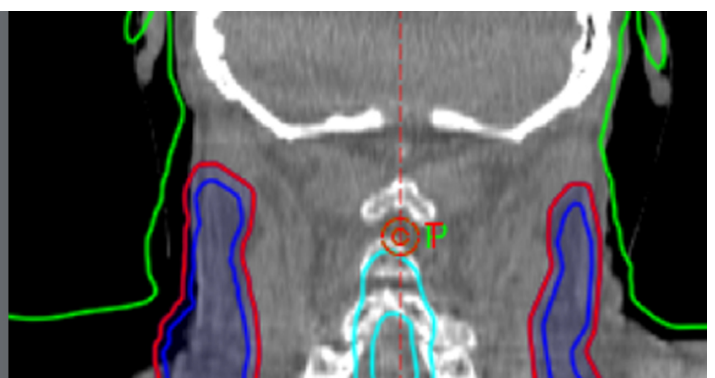
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An assessment of the efficiency of methods for measurement of the computed tomography dose index (CTDI) for cone beam (CBCT) dosimetry by Monte Carlo simulation

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Abstract

The IEC has introduced a practical approach to overcome shortcomings of the CTDI₁₀₀ for measurements on wide beams employed for cone beam (CBCT) scans. This study evaluated the efficiency of this approach (CTDI_{IEC}) for different arrangements using Monte Carlo simulation techniques, and compared CTDI_{IEC} to the efficiency of CTDI₁₀₀ for CBCT. Monte Carlo EGSnrc/BEAMnrc and EGSnrc/DOSXYZnrc codes were used to simulate the kV imaging system mounted on a Varian TrueBeam linear accelerator. The Monte Carlo model was benchmarked against experimental measurements and good agreement shown. Standard PMMA head and body phantoms with lengths 150, 600, and 900 mm were simulated. Beam widths studied ranged from 20–300 mm, and four scanning protocols using two acquisition modes were utilized. The efficiency values were calculated at the centre (ϵ_c) and periphery (ϵ_p) of the phantoms and for the weighted CTDI (ϵ_w). The efficiency values for CTDI₁₀₀ were approximately constant for beam widths 20–40 mm, where $\epsilon_c(\text{CTDI}_{100})$, $\epsilon_p(\text{CTDI}_{100})$, and $\epsilon_w(\text{CTDI}_{100})$ were $74.7 \pm 0.6\%$, $84.6 \pm 0.3\%$, and $80.9 \pm 0.4\%$, for the head phantom and $59.7 \pm 0.3\%$,

$82.1 \pm 0.3\%$, and $74.9 \pm 0.3\%$, for the body phantom, respectively. When beam width increased beyond 40 mm, $\epsilon(\text{CTDI}_{100})$ values fell steadily reaching $\sim 30\%$ at a beam width of 300 mm. In contrast, the efficiency of the CTDI_{IEC} was approximately constant over all beam widths, demonstrating its suitability for assessment of CBCT. $\epsilon_c(\text{CTDI}_{\text{IEC}})$, $\epsilon_p(\text{CTDI}_{\text{IEC}})$, and $\epsilon_w(\text{CTDI}_{\text{IEC}})$ were $76.1 \pm 0.9\%$, $85.9 \pm 1.0\%$, and $82.2 \pm 0.9\%$ for the head phantom and $60.6 \pm 0.7\%$, $82.8 \pm 0.8\%$, and $75.8 \pm 0.7\%$, for the body phantom, respectively, within 2% of $\epsilon(\text{CTDI}_{100})$ values for narrower beam widths. $\text{CTDI}_{100,w}$ and $\text{CTDI}_{\text{IEC},w}$ underestimate $\text{CTDI}_{\infty,w}$ by $\sim 55\%$ and $\sim 18\%$ for the head phantom and by $\sim 56\%$ and $\sim 24\%$ for the body phantom, respectively, using a clinical beam width 198 mm. The CTDI_{IEC} approach addresses the dependency of efficiency on beam width successfully and correction factors have been derived to allow calculation of CTDI_{∞} .

Keywords: Monte Carlo, EGSnrc/BEAMnrc, EGSnrc/DOSXYZnrc, CTDI, CBCT, efficiency, IEC

(Some figures may appear in colour only in the online journal)

1. Introduction

Computed tomography (CT) is an important imaging procedure, but the radiation dose required is relatively high compared to conventional radiographic procedures. Therefore radiation protection should be optimised to minimise risks of adverse health effects (Brenner and Hall 2007, Smith-Bindman *et al* 2009, Brenner *et al* 2011, Pearce *et al* 2012). It is essential that performance and output of CT scanners are tested routinely. The main dose performance descriptor for CT scanning, known as the CT dose index (CTDI), aims to quantify the dose associated with the region being scanned (Shope *et al* 1981). The CTDI was originally proposed for single CT slices to estimate the multiple scan average dose (MSAD), but since its introduction a number of new techniques have been developed, such as multi-slice CT, helical scanning, and cone beam CT (CBCT). These developments brought the need to adapt the CTDI concept if it was to continue to be utilized effectively. Various dose descriptors were derived based on the CTDI concept namely CTDI_{FDA} , CTDI_{100} , weighted CTDI_w , volume CTDI_{vol} , and the dose length product (DLP) (McNitt-Gray 2002). The CTDI_{100} concept has been developed to measure the useful output from a single beam rotation with a 100 mm long pencil ionization chamber located at the isocentre, parallel to the scanner axis (IEC 2001). Measurements with a similar chamber are also made inside standard head and body phantoms made of polymethyl methacrylate (PMMA), which are 150 cm long and have diameters of 16 and 32 cm respectively, in order to provide data that can be related to patient doses. The descriptors CTDI_w , CTDI_{vol} , and DLP are based on assessments of the CTDI_{100} within such phantoms. However, it has been shown that these types of CTDI_{100} measurement have a major shortcoming, particularly for wide CT beams such as those employed for CBCT scans in that they do not capture all the radiation that would contribute to the patient dose (Mori *et al* 2005, Boone 2007, Kyriakou *et al* 2008, Geleijns *et al* 2009). This shortcoming brought the need for further adaptation of the CTDI_{100} concept. These adapted methods are relevant to CBCT scans that are obtained with wide beams and a single rotation in most cases. Different approaches to overcome the CTDI_{100} limitations have been proposed by the International Electrotechnical Commission (IEC) (IEC 2009) and the American Association of Physicists in Medicine (AAPM) task group (111) (AAPM 2010). The IEC approach involves making measurements

with a narrower reference beam and adjusting these based on a ratio of CTDI_{100} measurements in air. This approach has been recommended by the International Atomic Energy Agency (IAEA) (IAEA 2011) and the Institute of Physics and Engineering in Medicine (IPEM) (Platten *et al* 2013). The AAPM approach is based on measuring the cumulative dose with a small ion chamber in the centre of a phantom that is sufficiently long to provide an equilibrium condition for measurement of scatter radiation, such that any further increase in the length of the phantom will not increase the measured dose.

The efficiency of the CTDI_{100} technique is assessed as the ratio of the CTDI_{100} measurement in a standard phantom to that recorded by an infinitely long chamber within an infinitely long PMMA phantom, which equates to the total dose received by a patient undergoing a CT scan. This is considered to be a good indicator for ability of the CTDI_{100} to report the actual dose absorbed by the patient. The aim of this study was to evaluate the efficiency of the new IEC dosimetry arrangement using Monte Carlo simulation techniques and compare it to the efficiency of CTDI_{100} . The influence of the reference beam width on the efficiency calculation was also investigated. Results were analysed and compared with experimental measurements for three scanning protocols (head, thorax, and pelvis) frequently used in the clinic.

2. The dosimetry approaches

The CTDI_{100} is defined as:

$$\text{CTDI}_{100} = \frac{1}{N \times T} \int_{-50 \text{ mm}}^{50 \text{ mm}} D(z) \, dz \quad (1)$$

where N is the number of slices acquired in a single scan, T is the nominal thickness of one slice, and $D(z)$ is the air kerma at position z from a single axial rotation at $z = 0$. Measurements of CTDI_{100} in a phantom are made at the centre and at four peripheral positions 1 cm below the phantom surface. To take account of heterogeneity in the air kerma distribution within the phantom, a weighted CTDI_{100} is defined as:

$$\text{CTDI}_w = \frac{1}{3} \text{CTDI}_{100,c} + \frac{2}{3} \text{CTDI}_{100,p} \quad (2)$$

Where $\text{CTDI}_{100,c}$ is measured at the central hole in the phantom and $\text{CTDI}_{100,p}$ is calculated as an average of four measurements made at the peripheral positions.

A major shortcoming of the CTDI_{100} for wide beams is that it does not capture all the radiation that would contribute to the patient dose. The efficiency (ϵ) equal to the ratio of CTDI_{100} measured within standard PMMA phantoms to that measured with an infinitely long chamber within infinitely long PMMA phantoms (CTDI_{∞}) can be used as an indicator of the ability of CTDI_{100} to record a value representative of the radiation exposure.

$$\epsilon = \frac{\text{CTDI}_{100}}{\text{CTDI}_{\infty}} \quad (3)$$

where CTDI_{∞} is equal to:

$$\text{CTDI}_{\infty} = \frac{1}{N \times T} \int_{-\infty}^{\infty} D(z) \, dz \quad (4)$$

The IEC approach (CTDI_{IEC}) is to be employed for beams of width > 40 mm and involves the application of a correction factor to measurements of CTDI₁₀₀ for a narrow reference CT beam. The correction factor is a ratio of two CTDI measurements free in air (FIA) for the beam width of interest (CTDI_{FIA, (N × T)}) and the reference beam width (CTDI_{FIA, ref}), and is defined as follows:

$$\text{CTDI}_{\text{IEC}} = \text{CTDI}_{100, \text{ref}} \times \left(\frac{\text{CTDI}_{\text{FIA}, (N \times T)}}{\text{CTDI}_{\text{FIA}, \text{ref}}} \right) \quad (5)$$

where CTDI_{100, ref} is the CTDI₁₀₀ of a narrow reference beam width (≤ 40 mm) measured within the standard PMMA phantoms and CTDI_{FIA} is equal to:

$$\text{CTDI}_{\text{FIA}} = \frac{1}{N \times T} \int_{-L/2}^{L/2} D(z) \, dz \quad (6)$$

where L is the minimum integration length. This depends on $(N \times T)$; (1) if $N \times T \leq 60$ mm then $L = 100$ mm, or (2) if $N \times T > 60$ mm, then $L = N \times T + 40$ mm. For the beam widths that fall within the first range, measurements of CTDI_{FIA} are made in a manner similar to that for the standard CTDI₁₀₀, but for wider beams, a longer ionization chamber that covers $L = N \times T + 40$ mm is required. If a long ionization chamber is not available, a standard 100 mm ionization chamber can be used by implementing a practical technique (IAEA 2011). This technique involves moving the 100 mm ionization chamber across the entire beam width in 100 mm steps to achieve the required L . Utilizing this technique integration lengths are multiples of 100 mm (i.e. 200, 300, etc). The sum of the dose resulting from these steps is then multiplied by the ionization chamber length 100 mm and divided by the width of the beam of interest to calculate CTDI_{FIA}.

3. Materials and methods

3.1. The kV-system and the scanning protocols

This study involved the kV imaging system mounted on a Varian Truebeam linear accelerator (Varian Medical systems, Palo Alto, CA). All the imaging modes available in the system namely radiography, fluoroscopy and CBCT mode were utilized. The CBCT mode acquires a 3D image formed from a large number of radiographic projections acquired during rotation of the kV source around the patient over arcs of 360° or 200°. Either of two acquisition modes: the Full-Fan and the Half-Fan can be used depending on the size of the scanned target region. For a smaller target such as the head, the Full-Fan mode is employed with a full bowtie filter, and for larger regions such as the chest or pelvis, the Half-Fan mode is employed with a half bowtie filter (figure 1). The Full-Fan mode scans the target symmetrically with a field of view (FOV) of 26.4 cm at the isocentre. The collimator blades X_1 and X_2 that control the FOV are set to 13.2 cm, and the kV source and the flat plane detector are rotated simultaneously around a patient lying on a stationary table in axial mode to acquire a volumetric image for the target as shown in figures 1(a) and (b). The Half-Fan mode allows a larger FOV by scanning target volumes asymmetrically, where the X_1 and X_2 are set to 2.6 cm and 23.9 cm, and the flat plane detector is shifted laterally by 14.8 cm to acquire a FOV of 47.8 cm at the isocentre as shown in figures 1(c) and (d). The system allows selecting any field size between (2 cm × 2 cm) and (50 cm × 50 cm) at the isocentre. Two focal spots of size 0.4 mm used for the radiography and fluoroscopy modes and 1.0 mm used for the CBCT mode were used. Four scanning

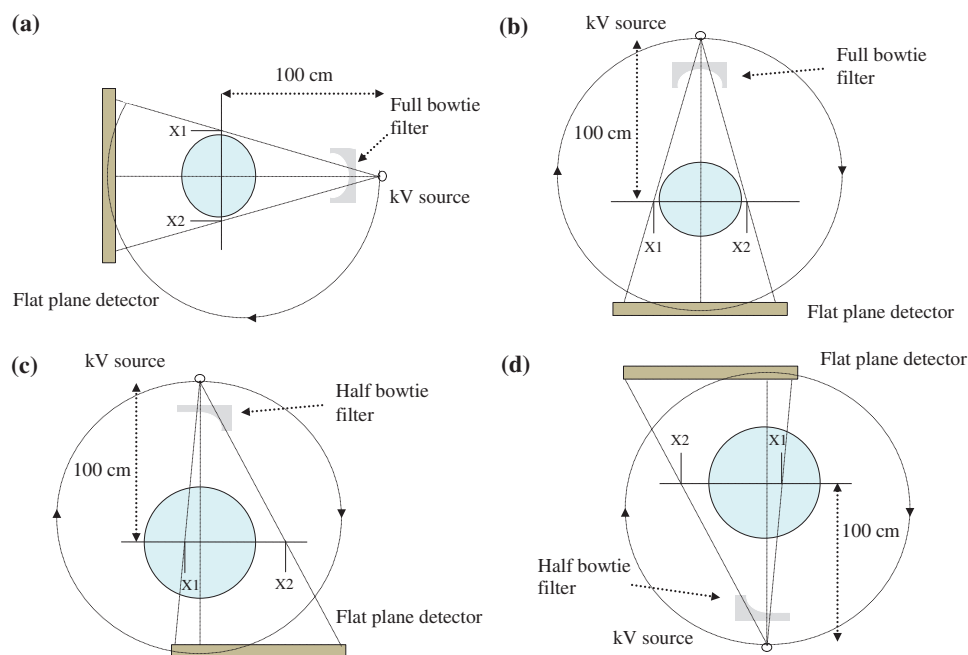


Figure 1. Diagrammatic representations of the Truebeam scanning modes. (a, b) Full-Fan mode employed for scanning smaller targets such as the head through (a) a partial scan (200°; 90° to 290°) and (b) a full scan (360°). (c, d) Half-Fan mode used for scans of the trunk through a full scan (360°), with c and d each covering the two halves of the scan. Values of the lateral collimator blades X1 and X2 are shown in table 1.

protocols listed in table 1, namely Head-200 (figure 1(a)), Head-360 (figure 1(b)), Body-200 (figure 1(a)), and Body-360 (figures 1(c) and (d)), were defined, which cover all scanning parameters used in the clinic. Head-200 and Head-360 are used for head scans, Body-200 covers the pelvic spot light scan that acquires an image for a target in the pelvic region posterior of a patient with the partial scan mode, and Body-360 covers thorax and pelvic scans.

3.2. Monte Carlo simulation

The Monte Carlo (MC) EGSnrc/BEAMnrc and EGSnrc/DOSXYZnrc codes (V4 2.4.0) released in 2013 were used in this study (Rogers *et al* 1995, Kawrakow and Rogers 2000, Walters *et al* 2013). EGSnrc code is an extended version of EGS4 code and had been initially developed to simulate radiotherapy applications (i.e. a linear accelerator) in the MV energy range. Subsequently, the code has been successfully benchmarked and improved to simulate kilovoltage applications such as CT and CBCT systems (Mainegra-Hing and Kawrakow 2006, Ali and Rogers 2007). In recent years, BEAMnrc and DOSXYZnrc codes have been extensively used to conduct dosimetric studies of CBCT systems (Downes *et al* 2009, Kim *et al* 2010, Deng *et al* 2012, Qiu *et al* 2012, Ding and Munro 2013). BEAMnrc code was used to model the kV system by using the materials and the geometrical specifications provided by the manufacturer. Pre-coded components known as component modules (CMs) were implemented in BEAMnrc to simplify modelling MV and kV applications with a high degree of accuracy (Rogers *et al* 1995, Rogers *et al* 2013). Each CM is utilized for a specific purpose, for example CMs known as XTUBE and SLABS are used to simulate an x-ray target and a filter of a specific

Table 1. The scanning protocols employed in this study. Body-360 is used with 20 mA, 264 mAs for thorax scans and with 80 mA, 1056 mAs for pelvic scans.

	Head-200	Head-360	Body-200	Body-360
X-ray voltage (kVp)	100	100	125	125
X-ray current (mA)	20	20	80	20, 80
X-ray millisecond (ms)	20	20	25	20
Exposures (mAs)	147	264	733	264, 1056
Acquisition mode	Full-Fan	Full-Fan	Full-Fan	Half-Fan
Scan Arc Gantry rotation (degrees)	200°	360°	200°	360°
left side- 90°, right side-270°	90°–290°	0°–360°	90°–290°	0°–360°
Bowtie filter	Full	Full	Full	Half
No of Projections	367	660	367	660
Longitudinal extent of beam (mm)	20–300			
Reconstruction FOV in lateral direction (mm)	264	264	264	478
Lateral collimator blade positions at the isocentre X1 and X2 (mm)	132, 132	132, 132	132, 132	25, 239
Axial collimator blade positions at the isocentre Y1 and Y2 (mm)	(10–150), (10–150)			

thickness and material, respectively. The users are required to have specifications for each part of the system, as each CM requires inputting various parameters such as materials and dimensions. Table 2 shows the CMs and materials used to model the kV system involved in this study.

All material data and cross sections used in this study were obtained from PEGS4 code. To improve efficiency of the simulations, several variance reduction techniques (VRTs) were used. Kawrakow *et al* (2004) compared the efficiency of three VRTs, directional bremsstrahlung splitting (DBS), selective bremsstrahlung splitting (SBS) and uniform bremsstrahlung splitting (UBS) techniques, and found that the DBS technique gave the highest efficiency for simulations. Mainegra-Hing and Kawrakow (2006) have also reported that the DBS technique increases the efficiency of an x-ray simulation for low kV energies up to 5–6 times compared to that for a simulation without a splitting technique and 60 times higher than a simulation with the UBS technique. Therefore, the DBS technique with a splitting number (NBRSP = 2×10^4) was employed in all simulations. ISOURC No. 10 of BEAMnrc code was selected which allows an electron beam with a specific diameter and angle to hit an x-ray target (anode) from the side. The electron beam was specified as a mono-energetic beam of energies 100 keV and 125 keV and a diameter of 0.4 mm or 1 mm, and the incident angle was 14°. The low energy thresholds for creation of secondary electrons (AE) and photons (AP) were set to 0.516 MeV and 0.001 MeV, and these values were also set to the cut-off energies for transport of the electrons (ECUT) and photons (PCUT). The National Institute of Standards and Technology (NIST) data were used for bremsstrahlung cross-sections, and Koch–Motz data were used for bremsstrahlung angular sampling. The boundary tolerance parameter \$BDY_TOL was changed from the default value of (1×10^{-5}) to (5×10^{-7}) to improve the MC calculations (Ali and Rogers 2008). Spin effect, the electron impact ionization, Rayleigh scattering, and atomic relaxations were included in all simulations (Ding *et al* 2007).

3.3. Validation of the Monte Carlo model

In order to validate the MC model, a number of experimental measurements namely x-ray spectra, half value layer (HVL), lateral (in-plane) and axial (cross-plane) dose profiles, and depth dose curves were compared against the MC results.

Table 2. The component modules and materials used to model the kV system involved in the study.

Component module	Component of the kV system	Material
XTUBE	x-ray target (anode)	95% Tungsten and 5% Rhodium
CONESTAK	Tube window and inherent filter	Beryllium and Aluminium
PYRAMIDS	Port	Lead
PYRAMIDS	Primary collimator	Lead
JAWS	Blades	Lead and Steel
SLABS	Beam hardening filter	Titanium
PYRAMIDS	Half and Full bowtie filters	Aluminium
SLABS	Housing window	Polycarbonate

3.3.1. X-ray spectra and beam quality. BEAMnrc code was used to run (8×10^8) histories for energies 100 kVp and 125 kVp with a field of size ($50 \times 50 \text{ cm}^2$) without the bowtie filters with the presence of the internal filters. Two phase space (PS) files were recorded at source-isocentre-distance (SID) = 100 cm and then analyzed by BEAM Data Processor (BEAMDP) code. The resulting x-ray spectra were compared against x-ray spectra obtained from a matlab-based code SPEKTR (V2.1) (Siewerdsen *et al* 2004). The SPEKTR code allows the user to generate an x-ray spectrum for a range of kV energies using filters of different materials and thicknesses.

To examine the beam quality for the designed MC model, the HVL measurements were made in terms of air kerma using two energies 100 kVp and 125 kVp. The radiographic mode which allows the system to be operated with a stationary source was used to deliver a beam with a field of size ($5 \text{ cm} \times 5 \text{ cm}$) without bowtie filter at SID = 100 cm. Different thicknesses of aluminium (Al) layers ranging from 0.5 to 5.5 mm were applied. The Al thicknesses were incremented in steps of 0.5 mm over the full range studied and by 0.1 mm or 0.2 mm in the region of the expected HVL. To minimize the effect of scattered radiation arising from the Al layers and the kV system housing, the Al layers were set up at a source-to-surface distance (SSD) = 65 cm, and a plane parallel ionization chamber (Fluke model: 96020 C (150 CC), US) with calibration traceable to a standard dosimetry laboratory was placed at SID = 100 cm. An exponential curve from the measurements for each energy was obtained from which the HVL was estimated.

The HVL calculation for the MC model was accomplished by calculating the dose in an air cavity using CAVRZnrc code (Ding *et al* 2007, Kim *et al* 2012). The plane parallel ionization chamber used in the experimental measurements was modelled in CAVRZnrc using specifications of the chamber obtained from the manufacturer. 10^9 histories were run in BEAMnrc to record PS files for each energy at SSD = 65 cm, which were subsequently used as sources in CAVRZnrc. Al layers of thicknesses 0.5–6 mm with an increment of 0.5 mm were utilized to calculate the HVL value. The designed chamber was placed at SID = 100 cm, and the dose for each Al thickness was calculated with a statistical uncertainty of less than 1%. The HVL value for each energy was estimated by fitting the resulting values obtained from CAVRZnrcn to exponential curves.

3.3.2. Dose profiles and depth dose curves measurements. The lateral (in-plane) and axial (cross-plane) dose profiles and depth dose curve measurements were used to validate the geometric design of the MC model. The dose profiles and the depth dose curves were measured by using two settings: 100 kVp with the full bowtie filter and 125 kVp with the half bowtie filter. The measurements were made in a ($30 \text{ cm} \times 30 \text{ cm} \times 30 \text{ cm}$) solid water phantom (Gammex, Middleton, WI, US) using a Gafchromic film (International Specialty Products—Advanced Materials, New Jersey, US) XR-QA calibrated as described in (Martin *et al* 2011). The film

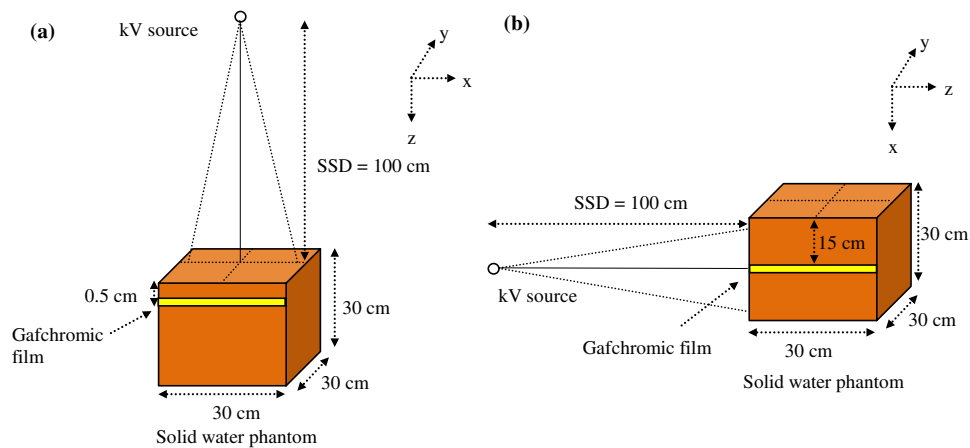


Figure 2. The experimental set up for (a) dose profiles measurements, and (b) depth dose curve measurements.

sheet was cut into strips of size 30 cm $\times$ 3 cm and 15 cm $\times$ 3 cm. The fluoroscopy mode was employed for all the measurements to deliver a continuous beam for 5–6 min to minimize fluctuations that may arise from the different responses of film from low exposure times. For the dose profiles, the solid water phantom was set up at SSD = 100 cm, and the strips were placed at a depth of 0.5 cm inside the phantom along x-axis for the lateral profiles and along y-axis for the axial profiles (figure 2(a)). In order to ensure that the beam widths used in the MC calculations were set up correctly, a lateral dose profile and four axial dose profiles of different beam widths were measured for each setting. A field of size 280 mm $\times$ 280 mm was used for the lateral dose profile, while 20 mm $\times$ 280 mm, 100 mm $\times$ 280 mm, 198 mm $\times$ 280 mm, and 280 $\times$ 280 mm were used for the axial dose profiles. For the depth dose curves, the kV source was moved 90°, and the strip was placed parallel to the beam at a depth of 15 cm inside the phantom along z-axis which was centred within a field of size 280 mm $\times$ 280 mm at SSD = 100 cm (figure 2(b)) (Fletcher and Mills 2008). The film strip of size 15 cm $\times$ 3 cm was employed for the beams of width 20 and 100 mm, and 30 cm $\times$ 3 cm for 280 and 198 mm and for the depth dose curve measurements.

The MC dose profiles and depth dose curves were derived by BEAMnrc and DOSXYZnrc codes using the experimental set up described in figure 2. First, 8 PS files (four for each setting) were recorded by BEAMnrc code at SID = 100 cm using 10^9 histories for each simulation. The field sizes used to generate the PS files were similar to those of the experimental measurements. Second, the PS files were used as sources in DOSXYZnrc code to calculate the absorbed dose inside the phantom by using (isource: 2). The solid water phantom used for the validation was designed as a cube of size (30 cm $\times$ 30 cm $\times$ 30 cm) and consisted of voxels of size (0.5 cm $\times$ 0.5 cm $\times$ 0.5 cm). In all the DOSXYZnrc simulations, 5×10^8 histories with a photon splitting number 300 were used to achieve an uncertainty of less than 2%. Walters and Kawrakow (2007) have shown that using the HOWFARLESS transport algorithm together with the PRESTA-I boundary crossing algorithm for a simulation carrying out with a homogeneous phantom and a PS file obtained from BEAMnrc code gave the highest efficiency for the dose calculation. Moreover, the HOWFARLESS transport algorithm was found to reduce the time required to complete a simulation significantly by accelerating transport of charged particle by ignoring voxel boundaries used for a homogeneous phantom and only considering distance of the charged particle to the outer boundaries (Walters and Kawrakow

2007). Therefore, both algorithms were applied to improve the efficiency of the calculations and minimize the simulation time. The output files were in (3ddose) format, which were analyzed subsequently by STATDOSE code.

3.4. Monte Carlo calculation for the efficiency

The efficiency values for the standard CTDI₁₀₀ and CTDI_{IEC} were evaluated and compared to demonstrate the improvement that can be gained by using the IEC approach CTDI_{IEC}. BEAMnrc and DOSXYZnrc codes were involved for the efficiency calculations. The efficiency of CTDI₁₀₀ was assessed at the centre (ϵ_c) and periphery (ϵ_p) of the phantom, and for CTDI_w (ϵ_w) as follows:

$$\epsilon_c(\text{CTDI}_{100}) = \frac{\text{CTDI}_{100,c}}{\text{CTDI}_{\infty,c}}, \epsilon_p(\text{CTDI}_{100}) = \frac{\text{CTDI}_{100,p}}{\text{CTDI}_{\infty,p}}, \epsilon_w(\text{CTDI}_{100}) = \frac{\text{CTDI}_w}{\text{CTDI}_{\infty,w}} \quad (7)$$

where CTDI _{∞,c} and CTDI _{∞,p} are calculated as in equation (4), and CTDI _{∞,w} is calculated as in equation (2). Similarly, the efficiency values for CTDI_{IEC} were assessed as follows:

$$\epsilon_c(\text{CTDI}_{\text{IEC}}) = \frac{\text{CTDI}_{\text{IEC},c}}{\text{CTDI}_{\infty,c}}, \epsilon_p(\text{CTDI}_{\text{IEC}}) = \frac{\text{CTDI}_{\text{IEC},p}}{\text{CTDI}_{\infty,p}}, \epsilon_w(\text{CTDI}_{\text{IEC}}) = \frac{\text{CTDI}_{\text{IEC},w}}{\text{CTDI}_{\infty,w}} \quad (8)$$

where CTDI_{IEC, c} and CTDI_{IEC, p} are calculated as in equation (5), and CTDI_{IEC, w} is calculated as in equation (2).

Mori *et al* (2005) and Kim *et al* (2011) have shown that the PMMA head and body phantoms of length 600 and 900 mm, respectively, are sufficient to provide the full scatter condition, thus these lengths were used to represent the infinite lengths for the head phantom $\left(\int_{-\infty}^{\infty} D(z) dz = \int_{-300}^{300} D(z) dz\right)$ and the body phantom $\left(\int_{-\infty}^{\infty} D(z) dz = \int_{-450}^{450} D(z) dz\right)$. The beam widths (20–300) mm and (60–300) mm with an increment of 20 mm were applied to assess $\epsilon(\text{CTDI}_{100})$ and $\epsilon(\text{CTDI}_{\text{IEC}})$, respectively. A beam of width 20 mm was used as the reference beam width for evaluating $\epsilon(\text{CTDI}_{\text{IEC}})$. The scanning protocols listed in table 1 were used to generate PS files using BEAMnrc code as described in section 3.2. 10^9 histories were run for each simulation, and PS files were recorded at SSD = 75 cm. Subsequently, the PS files were transferred into DOSXYZnrc and run with (isource: 8). The number of histories and the photon splitting number required to achieve uncertainties of 2–3% in DOSXYZnrc were different for the CTDI₁₀₀, CTDI_{FIA} and CTDI _{∞} quantities, thus the histories number ranged from (0.24×10^9) to (8×10^9) and the photon splitting number was (100–300). The efficiency values for $\epsilon(\text{CTDI}_{100})$ and $\epsilon(\text{CTDI}_{\text{IEC}})$ were estimated within two cylindrical phantoms of diameters 16 and 32 cm made of PMMA ($\rho = 1.19 \text{ g cm}^{-3}$) mm. In order to increase the speed of the DOSXYZnrc simulations, as the phantoms were made of a homogeneous medium, they were simulated with voxels of different sizes (Babcock *et al* 2008), smaller voxels being used at the centre and periphery of the phantoms and larger voxels for other areas. Lengths of the phantoms were 150, 600, and 900 mm, where 150 mm represented the standard length and the latter two were used for CTDI _{∞} . The CTDI_{FIA} was calculated using DOSXYZnrc code by simulating a pencil ionization chamber (Radcal, 20X6-3CT, US) whose length was equal to the integration length (L) described in equation (6). The HOWFARLESS transport algorithm and the PRESTA-I boundary crossing algorithm were employed for all simulations.

To indicate the influence of the reference beam width on $\epsilon(\text{CTDI}_{\text{IEC}})$, the efficiency values were calculated using another reference beam of width 40 mm. The influence was estimated

at the centre and periphery of the phantoms and for the weighted efficiency as the ratio of $\epsilon(\text{CTDI}_{\text{IEC}})$ calculated with 20 mm to those of 40 mm as follow:

$$\text{Ratio} = \frac{\epsilon(\text{CTDI}_{\text{IEC}})_{20}}{\epsilon(\text{CTDI}_{\text{IEC}})_{40}} \quad (9)$$

3.5. Validation of the Monte Carlo dose calculation

To ensure that the MC dose calculations and the designed phantoms used to calculate the $\epsilon(\text{CTDI}_{100})$ and $\epsilon(\text{CTDI}_{\text{IEC}})$ in section 3.4 were valid, the MC calculations for CTDI_{100} and CTDI_{FIA} , and dose profiles within the standard PMMA phantoms using the clinical beam width 198 mm were compared against experimental measurements.

The scanning protocols Head-200 and Body-360, which cover all parameters required to validate the MC calculations were used. The experimental measurements were carried out with a 100 mm pencil ionization chamber (Radcal, 20X6-3CT, US) with calibration traceable to a standard dosimetry laboratory. The CTDI_{100} values were measured at the central and peripheral positions using the 100 mm chamber within the standard PMMA phantoms placed at $\text{SID} = 100$ cm. The CTDI_{FIA} measurements were made using the technique described in section 2. An integration of length 300 mm was required to cover the beam width 198 mm, thus the chamber was moved three steps free in air at the isocentre (IAEA 2011). Additional experimental measurements for CTDI_{FIA} were made at the peripheral positions free in air to validate the MC calculations. The 100 mm ionization chamber was set up at 15 cm away from the isocentre for Body-360 and at 7 cm for Head-200. These measurements are not required for the QA procedure using the IEC approach, but they have been made to validate the MC calculations.

The experimental dose profiles were obtained at the centre of the standard PMMA phantoms for all the scanning protocols listed in table 1 using the beam width 198 mm. The phantoms were set up at $\text{SID} = 100$ cm, and four film strips of size (15 cm $\times$ 1 cm) (i.e. a film strip for each scanning protocol) were inserted into the central hole. In order to avoid an air gap resulting from removal of the central PMMA rod from the phantom, a PMMA rod was cut in half along its length so that the film strips could be inserted in a PMMA sandwich during the measurements. The resulting profiles were compared against MC dose profiles derived within PMMA phantoms of length 150 mm as described in sections 3.3.2 and 3.4.

3.6. CTDI measurements

CTDI_{100} and CTDI_{IEC} were assessed experimentally for three clinical scanning protocols (head, thorax, and pelvis). The clinical beam width 198 mm was used, and the protocol Head-200 was employed for the head scan, while Body-360 was used for the thorax and pelvic scans. Values for CTDI_{∞} for these protocols were estimated by the application of correction factors derived from the MC calculations obtained in section 3.4.

4. Results

4.1. Validation of the Monte Carlo model

Figure 3 shows a comparison between x-ray spectra for 100 kVp and 125 kVp generated by the MC model, that were obtained at $\text{SID} = 100$ cm with a field of size (50 cm $\times$ 50 cm), and the x-ray spectra generated by Spektr code. For both spectra, the number of photons at each energy was normalized with respect to the number of photons at the characteristic energy.

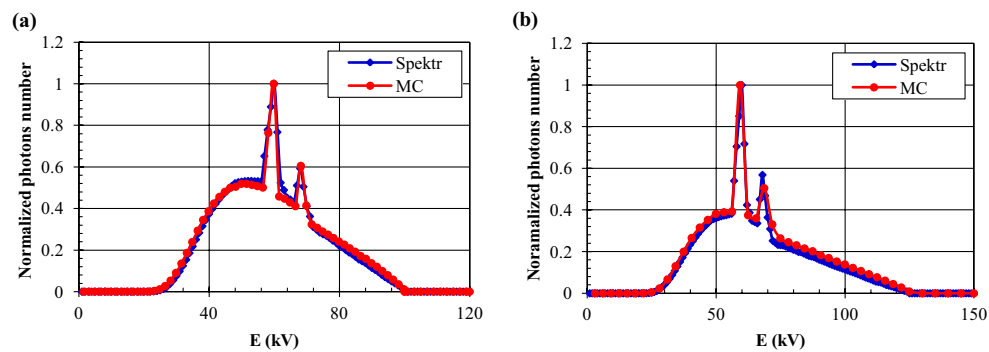


Figure 3. A comparison between the x-ray spectra of the MC model and Spektr, (a) 100 kVp and (b) 125 kVp.

Figure 4 shows a comparison between the MC and the experimental results for the HVL experiment. The HVL values calculated in MC were 3.84 mm Al and 4.89 mm Al for 100 kVp and 125 kVp, respectively. These values were within 0.05 mm of the respective experimental measurements 3.79 Al mm and 4.90 mm Al. The HVL values for the MC model agreed with the experimental measurements to within 1.30% for 100 kVp and 0.41% for 125 kVp.

The lateral and axial dose profiles and the depth dose curves obtained from the MC in terms of absorbed dose to the phantom showed good agreement with the experimental measurements (figure 5). Two settings were used: 100 kVp with the full bowtie filter (figures 5(a)–(c)), and 125 kVp with the half bowtie filter (figures 5(d)–(f)). The dose values for the profiles were normalized with respect to the central value, while those for the depth dose curves were normalised to the dose at depth 1 cm. The lateral dose profiles (figures 5(a) and (d)) were used to validate the geometrical design of the full and half bowtie filters, respectively. The lateral dose profiles of the MC for both settings matched to the experimental measurements to within $\pm 2.3\%$. (figures 5(b) and (e)) show the comparison between the axial dose profiles for different beam widths. The MC and the experimental axial profiles agreed within 3–5%, on average for both settings, but a larger variation was found at tails of the narrower beam width (20 mm) profiles. Such variations are likely to be caused by the accuracy of blade collimation for narrower beam widths (Varian 2010). The depth dose curves for the energies 100 kVp and 125 kVp obtained from the MC model (figures 5(c) and (f)) were within $\pm 2.2\%$ and $\pm 2.9\%$ of the experimental measurements, respectively.

4.2. Validation of the Monte Carlo dose calculation

Table 3 shows a comparison between CTDI_{100} and CTDI_{FIA} values measured experimentally using a beam of width 198 mm against those obtained from the MC. The results were normalised with respect to CTDI_{FIA} value at the isocentre for each protocol. The average variations between the MC and experimental values using the scanning protocols Body-360 and Head-200 were $\pm 2.9\%$ and $\pm 5.8\%$, respectively.

Figure 6 compares between dose profiles derived from the MC model and experimental profiles measured at the centre of the standard PMMA phantoms set up at $\text{SID} = 100$ cm. The dose values for all the profiles were normalized with respect to the central value. The dose profiles obtained with Head-200 and Head-360 (figures 6(a) and (b)) were within $\pm 1.7\%$ and $\pm 2.4\%$ of the experimental profiles, respectively, while Body-200 and Body-360 (figures 6(c) and (d)) were within $\pm 2.1\%$ and $\pm 2.5\%$, respectively.

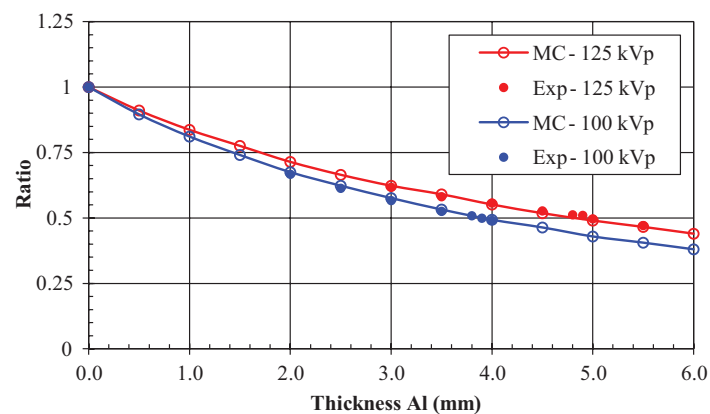


Figure 4. A comparison between the ratios of the dose in the air cavity with Al thicknesses (0.5–6.0) mm to the dose in the air cavity without a filter calculated with CAVR-Znrc and those measured experimentally (Exp). The fitted curves were used to estimate the HVL value for each energy. The MC calculations and the experimental measurements were carried out with a field of size (5 cm × 5 cm) and energies of 100 kVp and 125 kVp, and the chamber was placed at SID = 100 cm.

Table 3. A comparison between the MC values and experimental measurements (Exp) for CTDI₁₀₀ and CTDI_{FIA} for the scanning protocols Head-200 and Body-360 normalised with respect to the CTDI_{FIA} at the isocentre. p indicates measurements at the periphery.

	Body-360				Head-200			
	CTDI _{FIA}		CTDI ₁₀₀		CTDI _{FIA}		CTDI ₁₀₀	
	Exp		MC		Exp		MC	
centre	1.00	0.099	1.00	0.091	1.00	0.297	1.00	0.256
p (0°)	0.307	0.143	0.316	0.142	0.608	0.113	0.589	0.117
p (90°)	0.312	0.144	0.319	0.142	0.663	0.273	0.665	0.244
p (180°)	0.296	0.141	0.315	0.142	0.616	0.433	0.615	0.384
p (270°)	0.297	0.144	0.318	0.144	0.663	0.317	0.661	0.301
average p	0.302	0.143	0.317	0.143	0.637	0.284	0.632	0.261
CTDI _w	0.535	0.129	0.545	0.125	0.758	0.292	0.755	0.258

4.3. Calculation of the efficiency of CTDI₁₀₀ and CTDI_{IEC}

Figure 7 shows the efficiency values for the scanning protocols used in this study. $\epsilon(\text{CTDI}_{100})$ values for all protocols were approximately stable for the beam widths (20–40) mm. For Head-200 (figure 7(a)), values of $\epsilon(\text{CTDI}_{100})$, $\epsilon p(\text{CTDI}_{100})$, and $\epsilon w(\text{CTDI}_{100})$ for the beam widths (20–40) mm were $74.7 \pm 0.6\%$, $84.6 \pm 0.3\%$, and $80.9 \pm 0.4\%$, respectively, whereas for Head-360 (figure 7(b)) the values were $74.3 \pm 0.5\%$, $84.5 \pm 0.3\%$, and $80.7 \pm 0.4\%$, respectively. The efficiency values for the same beam widths for Body-200 (figure 7(c)) were $58.5 \pm 0.3\%$, $82.1 \pm 0.1\%$, and $74.7 \pm 0.2\%$, and for Body-360 (figure 7(d)) the values were $59.7 \pm 0.3\%$, $82.1 \pm 0.3\%$, and $74.9 \pm 0.3\%$, respectively. When the beam width increased beyond 40 mm, efficiency of all protocols fell steadily reaching values between ~30% and ~40% at the beam width 300 mm. However, $\epsilon(\text{CTDI}_{IEC})$ values for all the scanning protocols were within $\pm 1.0\%$ over all the beam widths (figure 7) (table 4). The beam

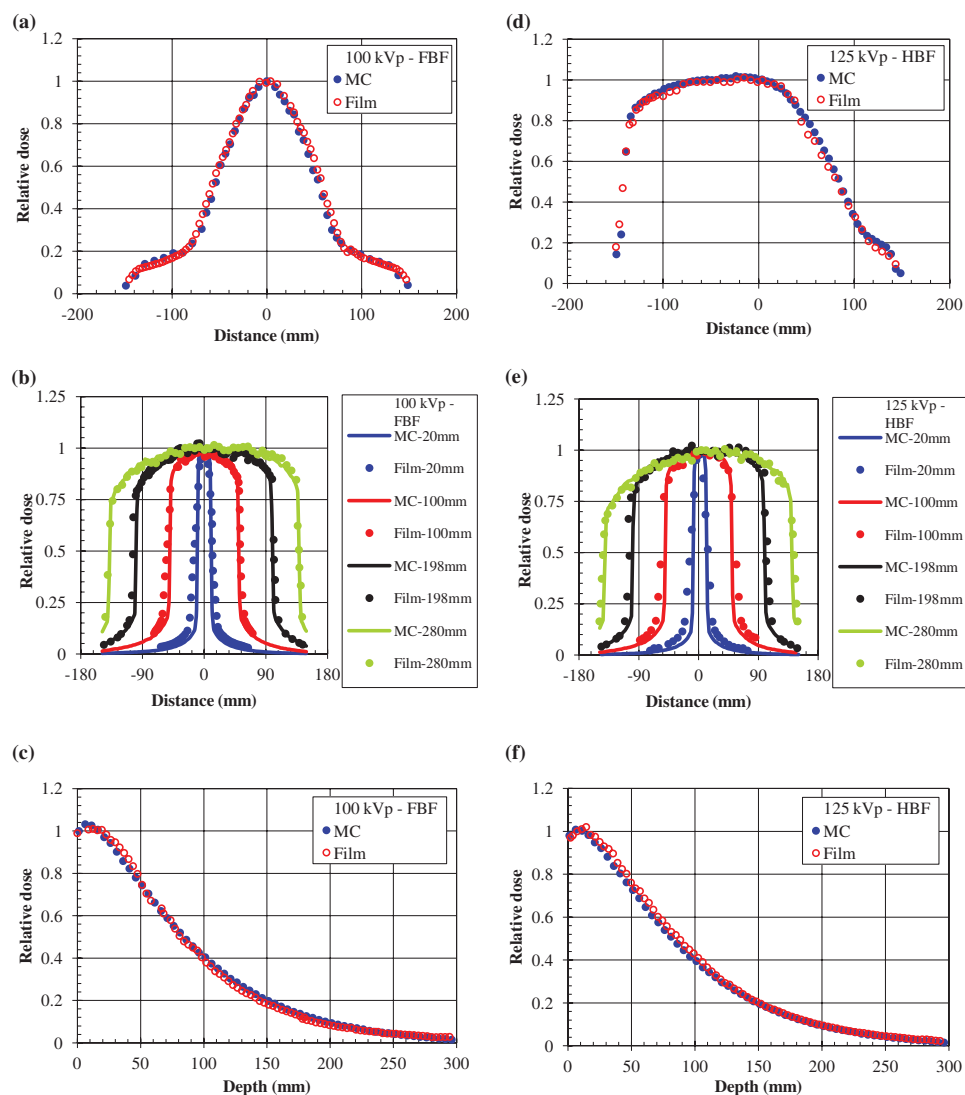


Figure 5. A comparison between dose profiles and depth dose curves of the MC and the experimental measurements made within a solid water phantom (Film). Settings used were (a–c) 125 kVp with the full bowtie filter (FBF), and (d–f) 100 kVp with the half bowtie filter (HBF). (a) and (d) the lateral dose profiles using the full bowtie and half bowtie filters, respectively, (b) and (e) the axial dose profiles for the beams of width 20, 100, 198, and 280 mm, (c) and (f) the depth dose curves.

width was found to play a minimal role in determining the efficiency values for the IEC approach (figure 7). Differences between the efficiency values calculated with half-scan 200° (figure 7(a)) and full-scan 360° (figure 7(b)), and full-fan (figure 7(c)) and half-fan (figure 7(d)) were within 1%. Therefore, $\epsilon(\text{CTDI}_{\text{IEC}})$ values were not affected by type of the scan or the bowtie filter significantly. Table 4 shows $\epsilon(\text{CTDI}_{\text{IEC}})$ values calculated with reference beams of width 20 and 40 mm. Differences were within $\pm 1\%$ for head scanning protocols, and $\pm 0.7\%$ for body protocols.

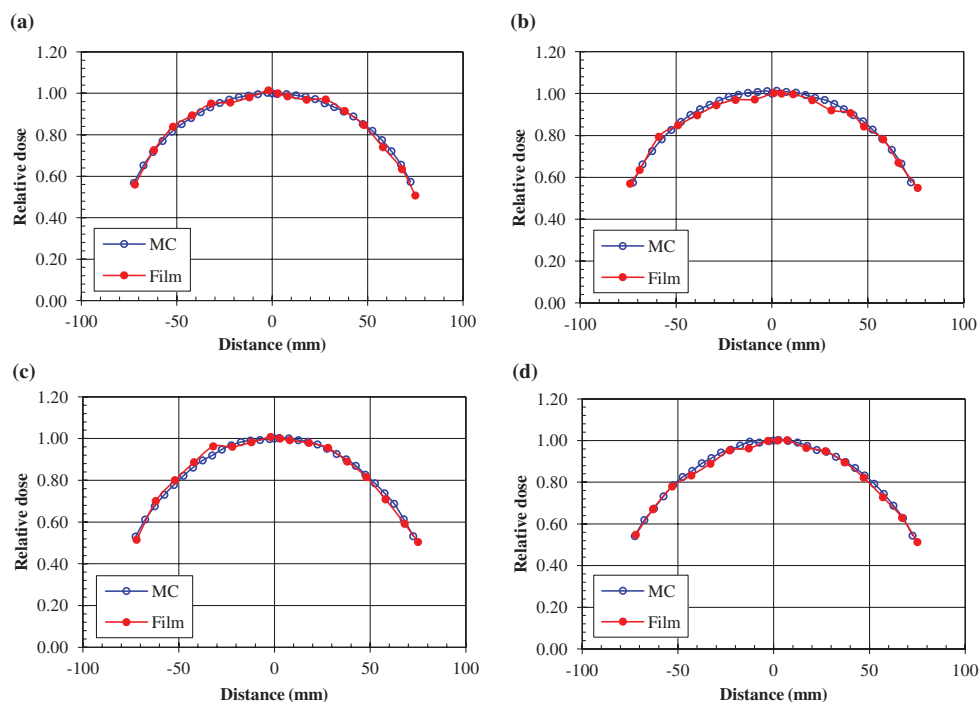


Figure 6. Comparison between dose profiles generated by MC simulation and those measured experimentally for the scanning protocols: (a) Head-200, (b) Head-360, (c) Body-200, and (d) Body-360.

4.4. CTDI measurements

The experimental values for CTDI_{100} were, as expected, lower than those of CTDI_{IEC} for all protocols (table 5), and agreed with the MC findings (figure 7). CTDI_{∞} values for the protocols (table 5) were estimated by application of correction factors (C_f) derived from $\varepsilon(\text{CTDI}_{\text{IEC}})_{20}$ values (table 4). These were used to estimate the efficiency of CTDI_{100} values measured experimentally (table 5) using equation (7). $\varepsilon_c(\text{CTDI}_{100})$, $\varepsilon_p(\text{CTDI}_{100})$ and $\varepsilon_w(\text{CTDI}_{100})$ for the head scan were 42.9%, 48.8%, and 46.6%, respectively, which agreed to within $\pm 1.7\%$ with the values derived from MC calculations of 41.5%, 47.1%, and 45.1%, respectively, using Head-200 (figure 7(a)). Similarly, the thorax and pelvic scans were within $\pm 0.9\%$ of the MC calculations, where $\varepsilon_c(\text{CTDI}_{100})$, $\varepsilon_p(\text{CTDI}_{100})$ and $\varepsilon_w(\text{CTDI}_{100})$ were 36.0%, 48.4%, and 44.6%, respectively for the thorax scan, and 35.8%, 48.0% and 44.6%, respectively, for the pelvic scan, and the MC calculations were 35.2%, 48.9%, and 44.5%, respectively using Body-360 (figure 7(d)).

5. Discussion

The efficiency of CTDI_{100} in collecting all the radiation to which a patient is exposed has become a more important issue since the introduction of advanced CT scanners using wider beams and CBCT. The efficiency of the CTDI_{100} has been studied extensively using a variety of techniques, beam widths and CT scanners (table 6). The MC results from the present study are in good agreement with other studies, and show that $\text{CTDI}_{100,w}$ underestimates the $\text{CTDI}_{\infty,w}$ by $\sim 18\%$ and $\sim 24\%$ for the head and body phantoms, respectively, using narrow

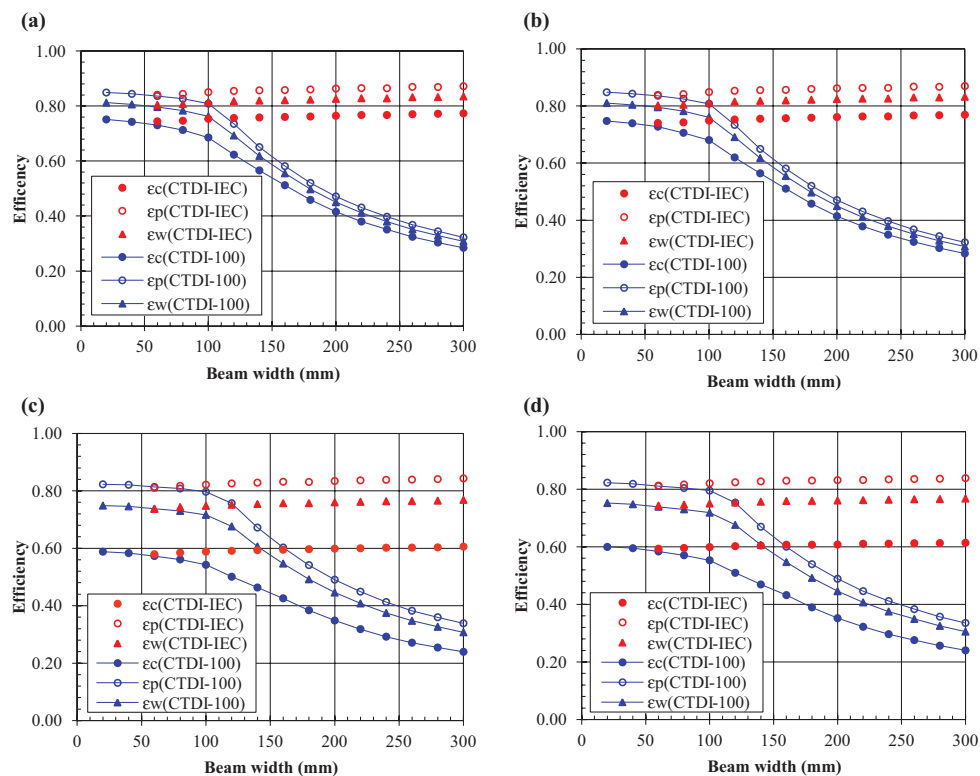


Figure 7. Efficiency values for CTDI₁₀₀ and CTDI_{IEC} calculated with MC using the four scanning protocols. Beam widths 20–300 mm were used for CTDI₁₀₀, while 60–300 mm were used for CTDI_{IEC}. The reference beam width was 20 mm. (a) Head-200, (b) Head-360, (c) Body-200, and (d) Body-360.

Table 4. The percentage $\epsilon(\text{CTDI}_{\text{IEC}})$ values at centre and periphery of the head and body phantoms, and for the weighted efficiency using the four scanning protocols and reference beams of width 20 and 40 mm.

	Head-200 (%)	Head-360 (%)	Body-200 (%)	Body-360 (%)
$\epsilon_c(\text{CTDI}_{\text{IEC}})_{20}$	76.1 ± 0.9	75.7 ± 0.9	59.5 ± 0.8	60.6 ± 0.7
$\epsilon_p(\text{CTDI}_{\text{IEC}})_{20}$	85.9 ± 1.0	85.7 ± 1.0	83.1 ± 1.0	82.8 ± 0.8
$\epsilon_w(\text{CTDI}_{\text{IEC}})_{20}$	82.2 ± 0.9	81.9 ± 1.0	75.6 ± 0.9	75.8 ± 0.7
$\epsilon_c(\text{CTDI}_{\text{IEC}})_{40}$	75.5 ± 0.9	75.2 ± 0.9	59.4 ± 0.8	60.3 ± 0.6
$\epsilon_p(\text{CTDI}_{\text{IEC}})_{40}$	85.7 ± 1.0	85.6 ± 1.0	83.5 ± 1.0	82.9 ± 0.8
$\epsilon_w(\text{CTDI}_{\text{IEC}})_{40}$	81.8 ± 0.9	81.7 ± 0.9	75.9 ± 0.9	75.8 ± 0.7

beam widths <40 mm. This underestimation increases for wider beams, where the primary beam extends beyond the length of the chamber and phantoms (figure 7). The variation of $\epsilon(\text{CTDI}_{100})$ with beam width found in the present study is similar to that reported for a CT scanner (Boone 2007) and a C-arm interventional CBCT scanner (Kyriakou *et al* 2008).

The IEC approach (CTDI_{IEC}) addresses the dependency of the efficiency on beam width successfully, with $\epsilon(\text{CTDI}_{\text{IEC}})$ values being essentially independent of beam width. Moreover, $\epsilon(\text{CTDI}_{\text{IEC}})$ values for the beam widths 60–300 mm were nearly identical to $\epsilon(\text{CTDI}_{100})$ values for narrow beam widths (figure 7, tables 4 and 6). Thus the CTDI_{IEC} successfully extends the application of the CTDI₁₀₀ approach to CBCT scans and provides a practical solution for the

Table 5. Experimental measurements for CTDI₁₀₀ and CTDI_{IEC} for the head, thorax, and pelvic scans using a beam of width 198 mm. The scanning protocol Head-200 with (20 mA, 147 mAs) was used for head scan, whereas Body-360 with (20 mA, 264 mAs) and (80 mA, 1056 mAs) was used for thorax and pelvic scans, respectively. The correction factors (C_f) within parentheses can be used to derive ($CTDI_{\infty} = C_f \times CTDI_{IEC}$).

	Head (mGy)			Thorax (mGy)			Pelvis (mGy)		
	CTDI ₁₀₀	CTDI _{IEC}	CTDI _∞	CTDI ₁₀₀	CTDI _{IEC}	CTDI _∞	CTDI ₁₀₀	CTDI _{IEC}	CTDI _∞
centre	1.72	3.06 (1.31)	4.01	1.73	2.95 (1.63)	4.81	6.87	11.79 (1.63)	19.21
average p	1.62	2.84 (1.17)	3.32	2.47	4.24 (1.20)	5.1	9.80	16.97 (1.20)	20.40
CTDI _w	1.66	2.92 (1.22)	3.56	2.21	3.81 (1.30)	4.95	8.84	15.25 (1.30)	19.82

use of CTDI₁₀₀ dosimetry equipment that are widely available in medical imaging and therapy departments for CBCT dosimetry.

The IAEA has recommended using a reference beam with a width of about 20 mm (IAEA 2011). Results from this study have shown that the difference between using 20 or 40 mm as the reference beam was not significant. However, the use of a 40 mm beam rather than 20 mm one is recommended for CBCT as these systems are manufactured for use with wide beams, and use of a narrower beam width may lead to uncertainty due to the accuracy of blade collimation (Varian 2010).

Although maintaining the CTDI₁₀₀ concept, the CTDI_{IEC} approach does not include all the radiation contributing to the patient exposure. This is inevitable with this approach unless a much longer phantom is used, which is less practical for routine measurement because of the requirement to handle the larger weight. Moreover, using a standard 100 mm ionization chamber requires the acquisition of a relatively large number of scans (nine) to estimate CTDI_{IEC,w} for one protocol using the practical technique described earlier. Moving the 100 mm ionization chamber using the practical technique over two or three positions increases the uncertainty of the measurements, which with three steps for a beam of width 160 mm was found to be 2% (Platten *et al* 2013). Therefore, use of a longer ionization chamber should be considered to minimize the number of scans and to reduce the uncertainty.

In addition to the underestimation associated with the IEC approach, the number of measurements required to complete a quality assurance (QA) procedure (i.e. extra time and effort) and the expected uncertainty, which may arise from the accuracy of blade collimation at a narrower beam and moving the ionization chamber over several steps, may limit the acceptability of this approach. As mentioned earlier, the AAPM has proposed another approach which requires replacing the widely available CT QA equipment (the standard ionization chamber and the PMMA phantoms) with a calibrated small ionization and longer phantoms. For CBCT scans, the number of measurements required for the AAPM approach is the same as for the CTDI₁₀₀ method. Also, the sources of uncertainty associated with the IEC approach are not present in the AAPM one. However, the use of larger phantoms may make implementation of the AAPM approach difficult for routine QA. The AAPM approach is at an early stage of development and the practical aspects are being investigated in the Task Group No. 200 (AAPM 2014). Both methods have advantages and disadvantages, and it might be hard for medical physicists to select the optimal method unless a universal dosimetry system for CBCT is agreed by all international organizations. The IEC approach recommended by the IAEA (IAEA 2011) and the IPEM (Platten *et al* 2013) is completely different from that proposed by the AAPM.

Several approaches have been proposed to improve $\varepsilon(CTDI_{100})$ for CBCT scans. Mori *et al* (2005) and Kyriakou *et al* (2008) have shown that the dose measurements need to be made with

Table 6. A comparison between $\epsilon(\text{CTDI}_{100})$ values in the present study obtained from the MC simulations and those from studies conducted with different techniques, beam widths, and CT scanners. MC: Monte Carlo, OSL: Optically Stimulated Luminescence, TLD: Thermoluminescent Dosimeter, Gaf. film: Gafchromic film, 20 mm IC: A Farmer type ionization chamber with an active length of 20 mm.

Investigator	Beam width (mm)	Phantom	x-ray voltage (kVp)	CT scanner	Technique	$\epsilon(\text{CTDI}_{100})$	$\epsilon_p(\text{CTDI}_{100})$	$\epsilon_w(\text{CTDI}_{100})$
(Li <i>et al</i> 2011)	18	Head	100	Siemens SOMATOM	MC	0.76	0.87	0.83
This study	20					0.75	0.85	0.81
(Li <i>et al</i> 2011)	40	Head	100	Siemens SOMATOM	MC	0.76	0.87	0.83
(Ruan <i>et al</i> 2010)	40					0.73	0.86	0.80
This study	40			GE LS VCT	OSL	0.74	0.84	0.80
(Li <i>et al</i> 2011)	18	Body	120	Siemens SOMATOM	MC	0.55	0.84	0.74
(Perinakis <i>et al</i> 2007)	18					0.63	0.78	
(Martin <i>et al</i> 2011)	20					0.57	0.83	
(Martin <i>et al</i> 2011)	20					0.55	0.81	
(Martin <i>et al</i> 2011)	20					0.60	0.85	
(Ruan <i>et al</i> 2010)	20					0.58	0.83	
(Boone 2007)	20					0.63	0.88	
(Dixon and Ballard 2007)	20					0.61	0.86	
(Dixon and Ballard 2007)	20					0.61	0.83	
(Mori <i>et al</i> 2005)	20							0.76
This study	20		125	Prototype 256	Photodiode	0.59	0.82	0.75
(Mori <i>et al</i> 2005)	42	Body	120	Prototype 256	Photodiode			0.74
(Martin <i>et al</i> 2011)	40					0.58	0.82	
(Li <i>et al</i> 2011)	40					0.55	0.83	0.74
(Ruan <i>et al</i> 2010)	40					0.56	0.81	
(Dixon and Ballard 2007)	40					0.61	0.81	
(Boone 2007)	40					0.62	0.87	
This study	40		125	GE LS	MC	0.59	0.82	0.74
(Mori <i>et al</i> 2005)	138	Body	120	Prototype 256	Photodiode			0.60
This study	140		125					0.61
(Geleijns <i>et al</i> 2009)	160	Body	120	Aquilion ONE	MC			0.59
This study	160		125					0.56

a long ionization chamber within longer phantoms to achieve an efficiency of more than 90%. However, using such equipment is likely to be impractical for routine measurements in the field and would require the replacement of existing dosimetry equipment. (Amer *et al* 2007) have proposed a cone beam dose index (CBDI) concept, where $(N \times T)$ given in equation (1) is replaced by a 100 mm length which represents the active length of a standard pencil ionization chamber. A similar concept for a CBDI called $\bar{D}_{100}$ has been investigated by (Geleijns *et al* 2009), who showed that the efficiency of $\bar{D}_{100}$ for a beam of width of 160 mm calculated within the standard PMMA phantoms was 89% and 95% for the standard head and body phantoms, respectively.

Our results together with other studies (Mori *et al* 2005, Boone 2007, Kyriakou *et al* 2008, Geleijns *et al* 2009) show that the $CTDI_{100}$ underestimates $CTDI_{\infty}$ significantly. For the clinical beam width 198 mm used in this study, $CTDI_{100,w}$ underestimates $CTDI_{\infty,w}$ by ~55% for head and by ~56% for body (figure 7). Therefore the continued use of the standard $CTDI_{100}$ concept for CBCT scans does not provide a good reflection of the relative doses involved in the CBCT imaging procedures, as only half of the scan dose received by a patient is measured. The efficiency, maintaining the use of the $CTDI_{100}$ equipment, and the simplicity (i.e. more practical for regular use in the medical environment) may be the more important aspects for the majority of medical physicists. The IEC approach provides a practical solution for the CBCT dosimetry that overcomes some of the $CTDI_{100}$ limitations for CBCT. The efficiency of the $CTDI_{IEC,w}$ is only ~82% for head and ~76% for body, but since $CTDI_{IEC}$ is independent of beam width (figure 7), this could be addressed by use of appropriate correction factors and values derived in this study for the clinical protocols used are given in table 4.

6. Conclusion

The MC BEAMnrc and DOSXYZnrc codes were used to evaluate the efficiency of $CTDI_{100}$ and $CTDI_{IEC}$ for the kV CBCT imaging system on a Truebeam linear accelerator for different scanning protocols and beam widths. The MC results were validated by comparison with experimental measurements which gave agreement within $\pm 3\%$. The IEC approach successfully extends the application of the $CTDI_{100}$ to CBCT scans and provides a practical solution using existing $CTDI_{100}$ dosimetry equipment. The simulations demonstrate that the efficiency $\epsilon(CTDI_{IEC})$ is essentially independent of beam width, with values of ~82% for head and ~76% for body, approximately equal to those for $\epsilon_w(CTDI_{100})$ at narrower beam widths. $CTDI_{IEC}$ has the same disadvantage as the $CTDI_{100,w}$ in that $CTDI_{\infty,w}$ is underestimated by ~18% for head scans and ~24% for body scans, but unlike the $CTDI_{100,w}$, the underestimation is independent of beam width. Correction factors that could be applied to $CTDI_{IEC}$ to overcome the underestimation of $CTDI_{\infty}$ have been derived from the MC calculations for clinical protocols used. Differences in values for $CTDI_{IEC}$ derived using beams with widths 20 mm and 40 mm as the reference were only $\pm 1\%$. The use of a 40 mm reference beam is recommended to reduce uncertainty associated with the accuracy of blade collimation.

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